

PSTAT 10 Data Science Principles

Lecture 10: Discrete Random Variables

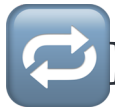
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Introduction



Review: Lecture 9

👉 Last lecture we looked at the fundamentals of probability, including:

- Events (outcomes)
- Sample Spaces
- Frequentist Probability
- Simulation
 - `sample()` Function
 - `replicate()` Function

Outline: Lecture 10

 This week we will be continuing our exploration of probability by studying:

- Types of events
- Conditional probability
- Discrete random variables
- Discrete Probability Distributions
 - Discrete Uniform
 - Bernoulli
 - Binomial
 - Poisson

Events

Events

DEFINITION

The **sample space** Ω is the collection of all **outcomes** ω of a random **experiment**.

- A **collection of outcomes** is known as an **event**.
- We typically denote events with capital letters, e.g. A , B or C .

Example:

A dice rolling experiment has **6 possible outcomes**, rolling a number 1 through 6.

- Our first event of interest might be to roll a six (i.e. one outcome).
- Our second event of interest might be to roll a number greater than 3 (i.e. the three outcomes 4, 5 and 6).

Certain and Probability-Zero Events

Clarifying Terminology

- We say that an event is **certain** (guaranteed, ensured, definitely going to happen) if it occurs **with probability 1**.
- Additionally, we say that an event occurs **with probability zero** if it “never” happens.



This is not the same as being impossible!

Example

- In the dice rolling experiment a **certain event** might be the event that we roll a number less than 7.
- An event that occurs **with probability zero** might be to roll a 7.
 - Okay in this case it's the same as being impossible, but not always!
 - In particular, when we consider continuous random variables, we will see that events can occur with probability zero but are not impossible.

! ? Probability Zero and Impossible

There is a subtle difference between an event that is **impossible** and an event that occurs **with probability zero**.

Infinite Bullseye

- To gain an intuitive understanding of the difference, consider a dartboard with area 2π (therefore with radius of 1).
 - We define an **infinitely small bullseye**, i.e. it is exactly the coordinate $(0, 0)$.
 - What is the probability a dart thrown uniformly at random exactly hits $(0, 0)$?
-
- The probability of hitting the bullseye is **zero** since the area of the bullseye is zero, but it is not **impossible** to hit the bullseye.
 - In fact, any point on the dartboard has probability zero of being hit, but it is not impossible to hit any point.
 - This is a little counter-intuitive!

Theoretical Dartboard



Dartboard with infinitely small bullseye

Example — Probability Zero Events

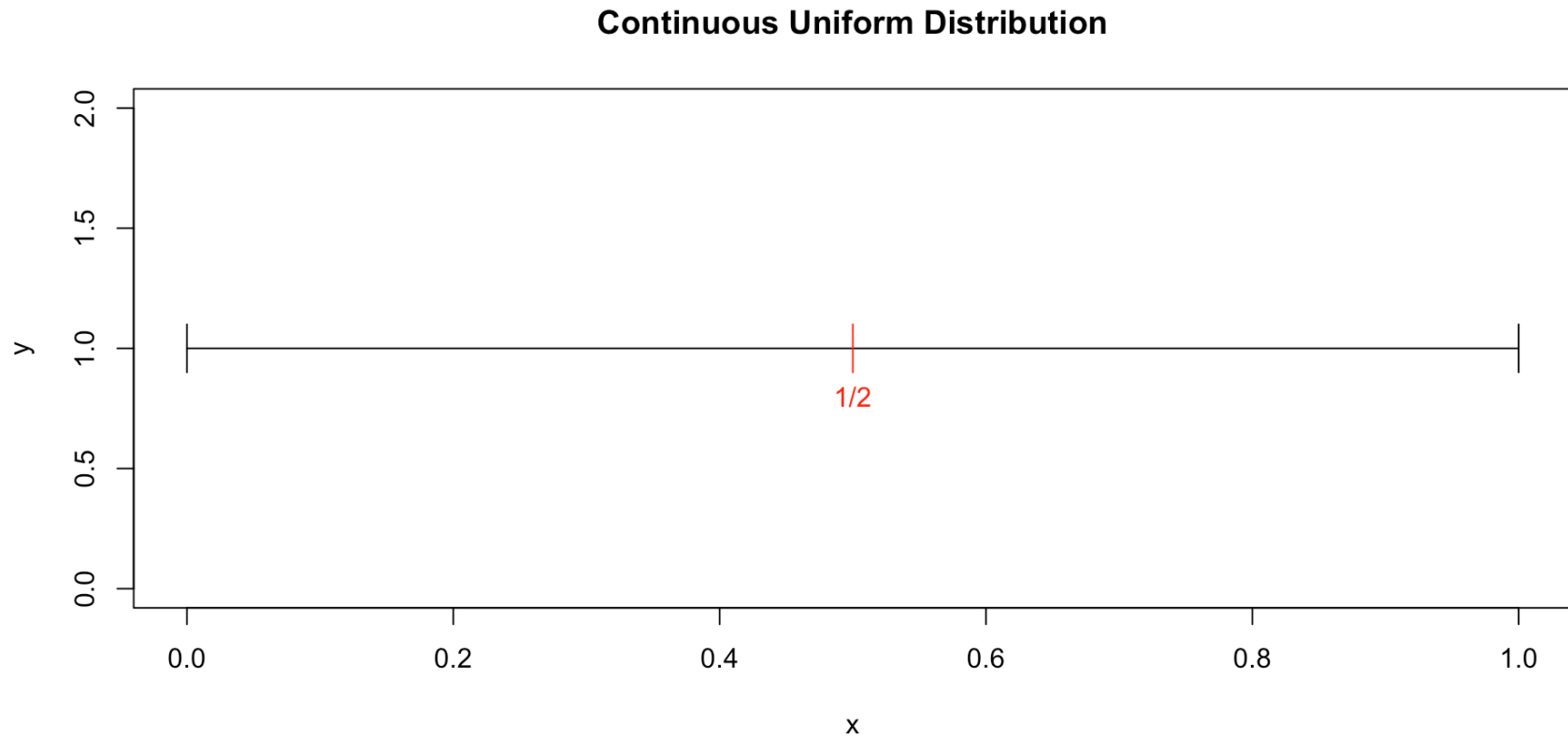
- Equivalently, consider the interval $[0, 1]$.
 - If we randomly select any number within the interval, what is the **probability that we pick exactly $1/2$** ?
 - Since there are **infinite numbers** in the interval between 0 and 1, the **probability** of selecting $1/2$ at random is 0.

!! This doesn't mean that it will never happen!

In fact, anytime we **randomly select a number** from a continuous distribution we are observing a probability zero event.



Visualizing a Probability-Zero Event



Selecting $1/2$ from the continuous interval from 0 to 1.

Probability Paradox

Lack of Intuition

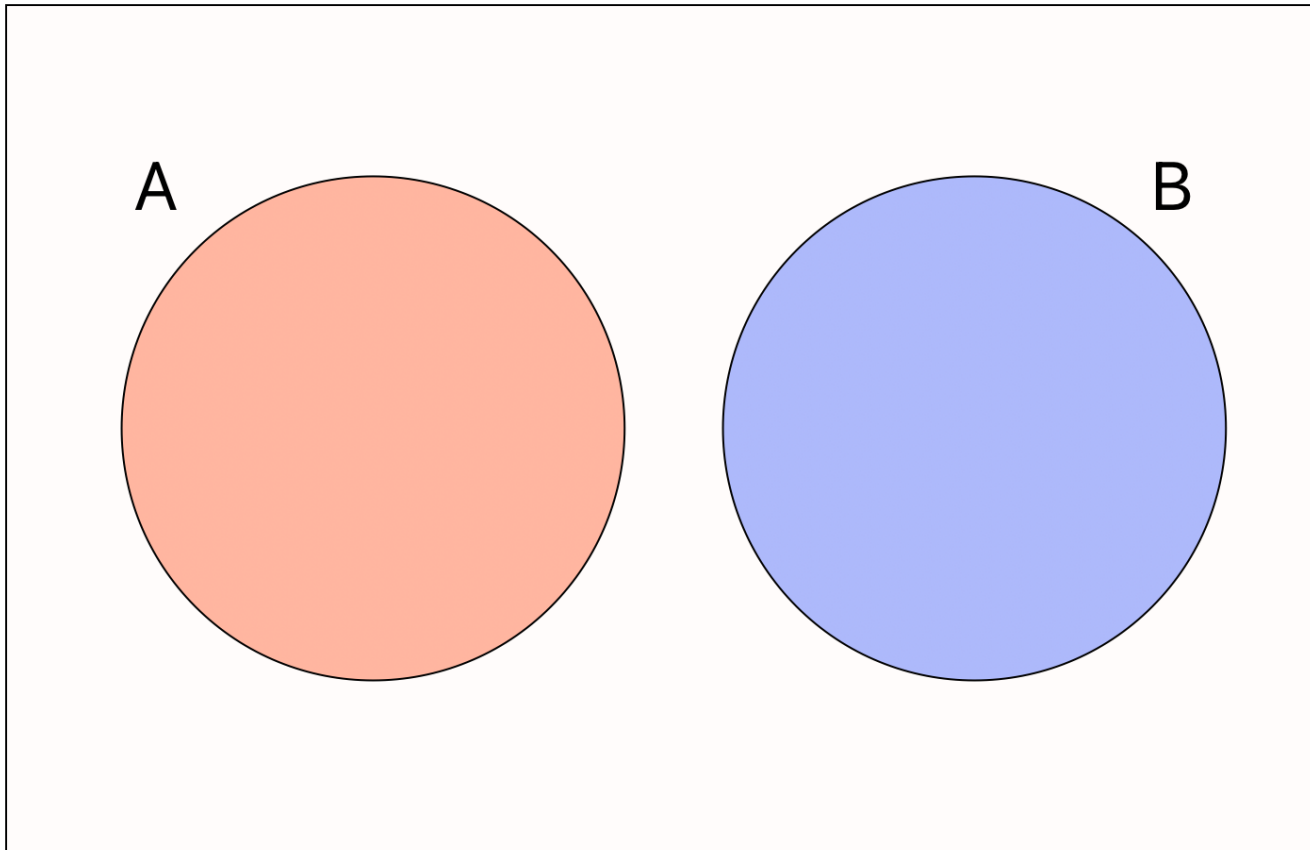
- Events with **probability zero can occur** — and do all the time.
 - The **odds** of them happening are still zero.
 - Intuition goes out the window.
- **Intuitive breakdowns** are common in many probability settings:
 - We can avoid (faulty) intuition with simulation.
 - Discrete logic simplifies things — computers figured this one out.

Let's now turn to the topic of **mutually exclusive** and **mutually inclusive** events.

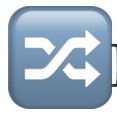
Mutual Exclusivity

We say that two events are **mutually exclusive** when

$$\mathbb{P}(A \cap B) = \mathbb{P}(A \text{ AND } B) = 0.$$



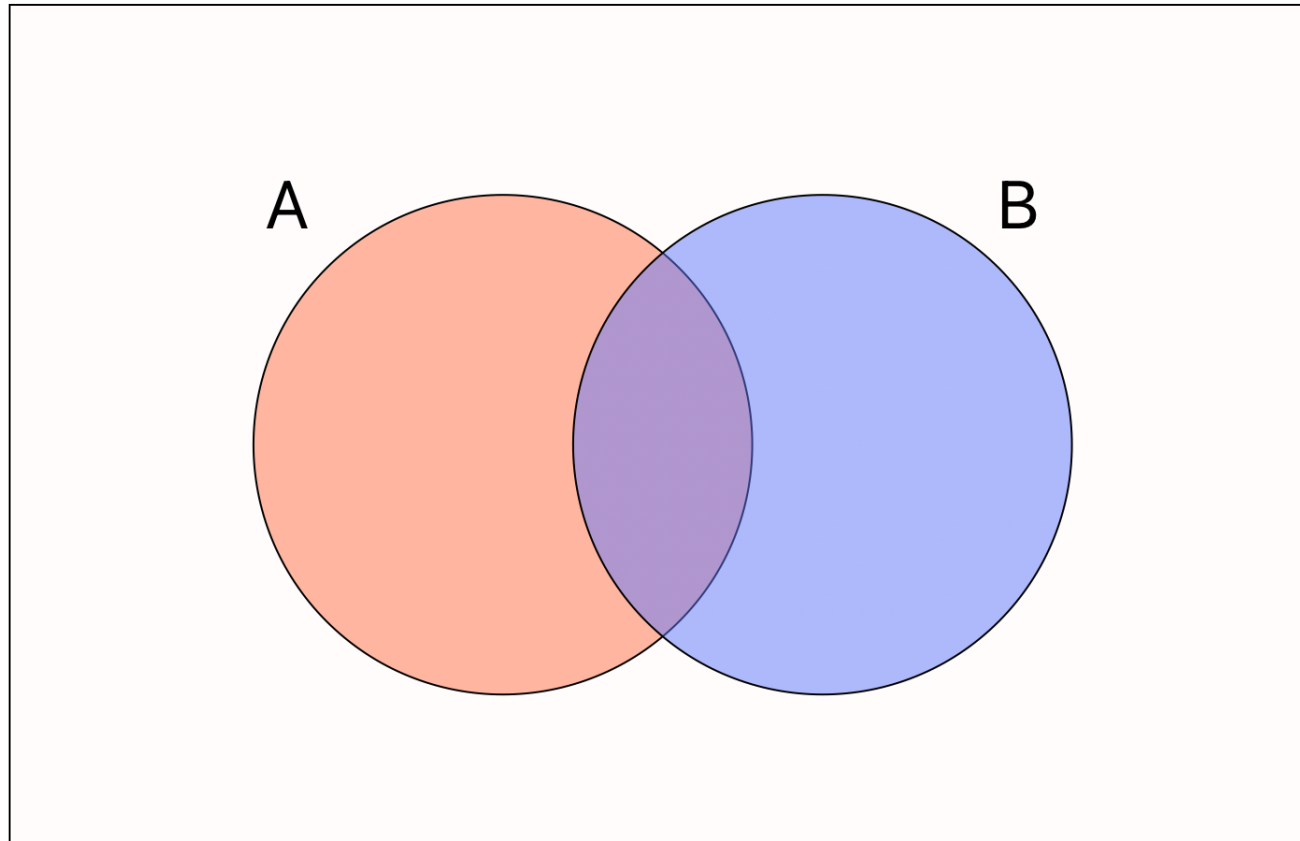
Mutually Exclusive Events



Mutual Inclusivity

When two events are **not mutually exclusive** we have that

$$\mathbb{P}(A \cap B) > 0.$$



Not Mutually Exclusive Events

+ Addition Rule

What is the addition rule?

DEFINITION

The **addition rule** in probability states that

$$\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B).$$

- Intuitively, the probability that event A or B occurs is equal to the **sum** of the probabilities that events A and B occur, **minus** the probability that both events A and B occur.
 - Can you see why we need to subtract the intersection?

Addition Rule — Mutual Exclusivity

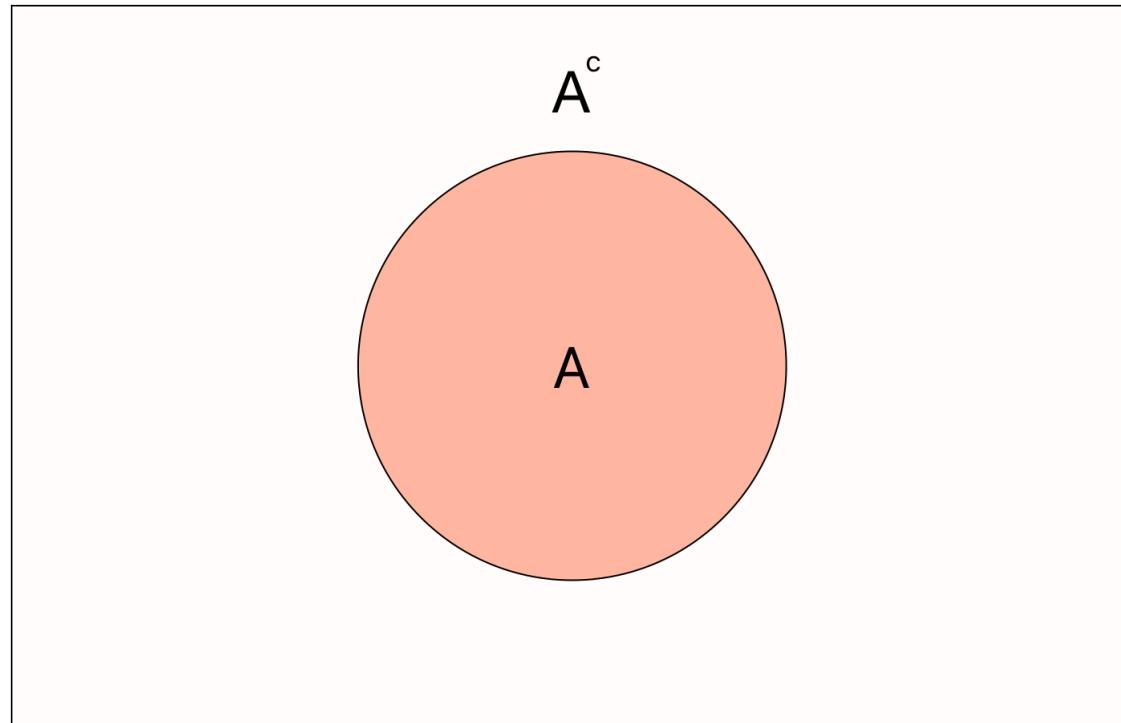
Recall that when A and B are **mutually exclusive** we have $\mathbb{P}(A \cap B) = 0$, and so for such events

$$\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B).$$

Complementary Events

The **complement** of an event A , written A^c , means all other outcomes besides those contained in A , i.e. the event such that

$$\mathbb{P}(A^c) = 1 - \mathbb{P}(A)$$



Complements

? Conditional Events

What is conditional probability?

Sometimes it is useful to think of the probability of an event A assuming some **condition** about a different event B .

DEFINITION

Formally, the **conditional probability** of event A **given** B is denoted

$$\mathbb{P}(A|B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}.$$

Examples

1. When rolling a 6-sided fair dice, what is the probability that we roll a number greater than 3 given that you roll an even number?
2. When drawing a card from a standard 52 card deck, what is the probability that you draw the ace of spades given that you draw an ace?

Independent Events

Mutual Exclusivity vs Independence

- **Mutually exclusive** events means that two events cannot happen at the same time.
- **Independence**, on the other hand, simply means that the outcome of one event **does not impact** the outcome of another.

DEFINITION

Formally, two events A and B are **independent** when:

$$\mathbb{P}(A|B) = \mathbb{P}(A),$$

- That is, when A conditioned on B occurs with the **same probability** as A unconditioned on B .



Bayes' Theorem

- It is often very **hard** to solve $\mathbb{P}(A|B)$ but very **easy** to solve $\mathbb{P}(B|A)$.
- In this case, **Bayes' Theorem** provides a useful identity for linking the two.

DEFINITION

Bayes' Theorem states that:

$$\mathbb{P}(A|B) = \frac{\mathbb{P}(B|A) \cdot \mathbb{P}(A)}{\mathbb{P}(B)}.$$

 Bayes' Theorem is particularly useful when it is **easy to simulate** $\mathbb{P}(A)$, $\mathbb{P}(B)$ and $\mathbb{P}(B|A)$, but very difficult to simulate $\mathbb{P}(A|B)$.



Exercise — Conditional Probability

Suppose you roll 3 fair 6-sided dice.

1. Write code using `m = 10000` simulations to approximate the probability that their **sum is at least 12 given that their product is at least 9**.

Make sure to set a seed using `set.seed()` — you might find the functions `apply()` and `which()` helpful.

04:00



Solution — Conditional Probability

```
1 set.seed(63864836)
2 n <- 10000
3 dice_rolls <- replicate(n, sample(1:6, 3, replace = TRUE))
4 # Not Using Bayes Rule
5 mean(apply(dice_rolls[, which(apply(dice_rolls, 2, prod) >= 9)], 2, sum) >= 12)
```

```
[1] 0.4459681
```

- Simulate `n = 10000` rolls of 3 dice, storing each roll as a column of `dice_rolls`.
- Use `which()` to find the columns (rolls) whose **product is at least 9**, then subset `dice_rolls` down to just those columns.
- Among that subset, use `apply()` and `mean()` to compute the **proportion whose sum is at least 12** — this directly approximates $\mathbb{P}(\text{sum} \geq 12 \mid \text{product} \geq 9)$.

```
1 # Using Bayes Rule
2 sum_rolls <- apply(dice_rolls, 2, sum)
3 prod_rolls <- apply(dice_rolls, 2, prod)
4 length(which(prod_rolls >= 9 & sum_rolls >= 12)) /
5   length(which(prod_rolls >= 9))
```

```
[1] 0.4459681
```

- Compute the sum and product of **every** roll across all `n` trials — no subsetting yet.
- Count how many rolls satisfy **both** conditions (`sum_rolls >= 12 & prod_rolls >= 9`) and divide by the count satisfying just the **conditioning event** (`prod_rolls >= 9`).
- This is the simulation version of $\mathbb{P}(A \mid B) = (\mathbb{P}(A \cap B)) / \mathbb{P}(B)$.

Discrete Random Variables

More Formal Concepts

In the next section we will introduce the following concepts:

- Discrete Random Variables
- Supports
- Probability Mass Functions
- Cumulative Distribution Functions

We will first quickly cover some **heuristic definitions** before considering several examples.

 Don't worry too much about the mathematics — focus on intuition!



Random Variables

What are random variables?

DEFINITION

A **random variable** is a mathematical object that can **take numerical values corresponding to outcomes** of a random experiment.

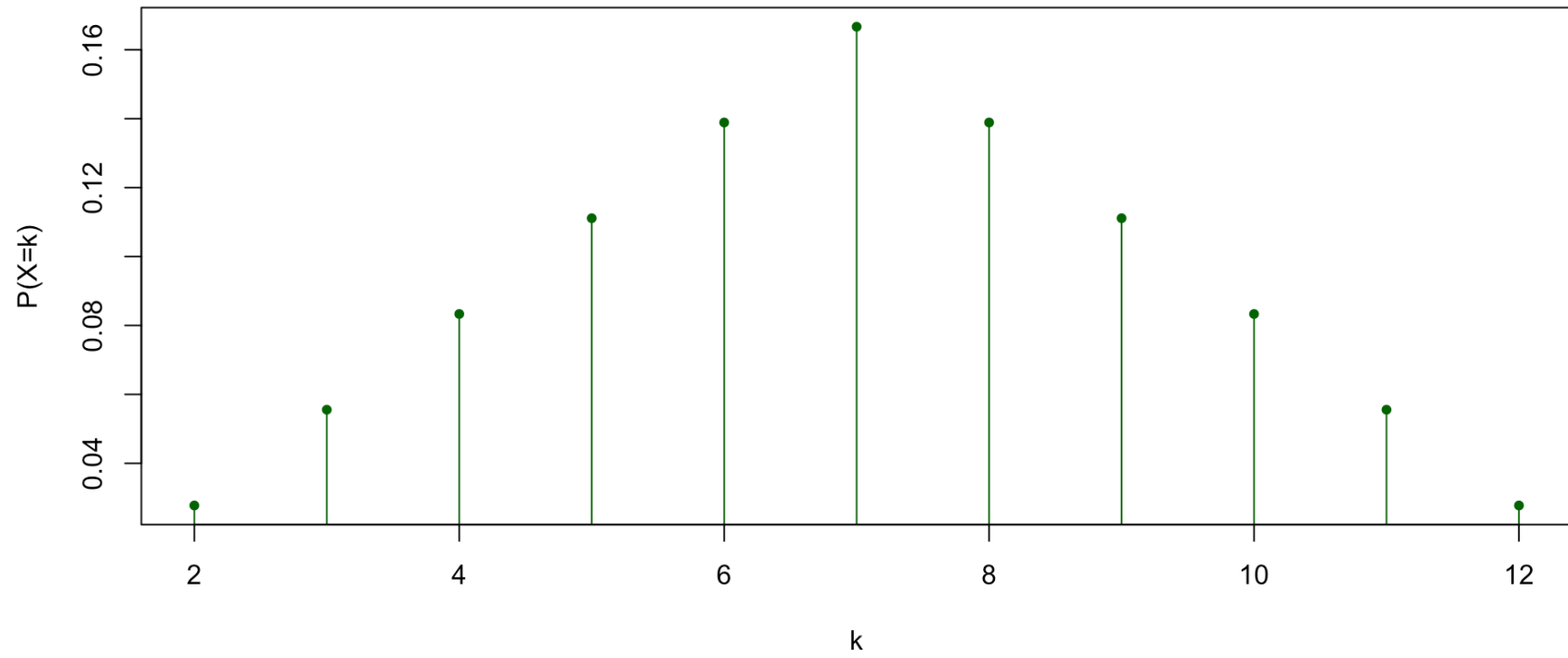
- We typically denote a **random variable** with capital letters, e.g. X , Y or Z , and their **realizations** (i.e. the values they take) by lowercase x , y or z .
- The range of values that each random variable X can take is known as the **support** of X .
- Intuitively, you can think of a random variable as a **function** that takes a value **randomly** according to some underlying probability.



Examples — Random Variables

- For coin flips we could define $X \in \{0, 1\}$ where $X = 1$ corresponds to heads and $X = 0$ tails.
- For fair 6-sided dice rolls we could define $X \in \{1, 2, \dots, 6\}$ where $X = 4$ means we rolled a 4.
- For the sum of dice rolls we could define $X \in \{2, 3, \dots, 12\}$.

Sum of Two Dice Rolls



Sum of Two Dice Rolls — Probability Mass Function

Discrete Random Variables

DEFINITION

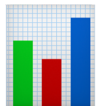
Each of the above random variables have a **discrete** (more specifically a countable) support, and are therefore called **discrete random variables**.

- Intuitively, a discrete random variable's possible values can be **listed out** one by one, even if the list is infinite (e.g. the counting numbers).
- Examples: the number of **heads in coin flips**, the **sum of dice rolls**, or the **number of customers** arriving in an hour.

DEFINITION

If a random variable can take (uncountably) **infinite** values (i.e. a continuous interval) we say that the random variable is **continuous**.

- Intuitively, a continuous random variable's possible values **cannot be listed out** — between any two possible values there are infinitely many more.
- Examples: a person's exact **height**, the **time** until a bus arrives, or a **temperature** reading.



Probability Mass Functions

DEFINITION

The **probability mass function (PMF)** of a **discrete random variable** X is a function $p_X(k) = \mathbb{P}(X = k)$ describing the **probability** of the random variable taking different values in its support.

Example:

Consider the dice roll experiment. The probability that X takes the values $1, \dots, 6$ are all $\frac{1}{6}$ and 0 for all other integers. We can write this

$$p_X(x) = \mathbb{P}(X = x) = \begin{cases} \frac{1}{6} & \text{if } x \in \{1, 2, \dots, 6\} \\ 0 & \text{otherwise.} \end{cases}$$

This is known as a **discrete uniform distribution**, one of several common named distributions.



Common Discrete Distributions

The following are some of the **most common probability distributions** for discrete random variables:

- Uniform Discrete Distribution
 - E.g. rolling a fair die, or drawing a random card rank.
- Bernoulli Distribution
 - E.g. a single coin flip, or whether a customer clicks an ad.
- Binomial Distribution
 - E.g. the number of heads in 10 coin flips, or the number of defective items in a batch.
- Poisson Distribution
 - E.g. the number of emails received per hour, or the number of earthquakes per year.



Uniform (Discrete) Distribution

DEFINITION

A **uniform (discrete) random variable** $X \sim \mathcal{U}(n)$ describes the outcome of an experiment with n **equally likely outcomes**.

Since we have n equal probabilities which must all add to one, the **probability mass function** of X is given by

$$p_X(k) = \frac{1}{n}; \quad k \in \{1, \dots, n\}$$



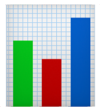
Often we do not specify that the **mass function is zero** for values other than the support and leave it implied.

We have already demonstrated how we use the `sample()` function for the discrete uniform.

Example: sampling a single value from $X \sim \mathcal{U}(6)$ (e.g. rolling one fair die):

```
1 sample(1:6, size = 1)
```

```
[1] 5
```



Long-Run Averages

Often we are not interested in the **probability of a specific event** but instead wish to know about the **average behaviour** of all events.

Let's consider the set of 12 counts of 2 summed dice.

```
1 set.seed(12)
2 rolls <- replicate(12, sum(sample(1:6, 2, replace = TRUE)))
3 rolls
4 mean(rolls)
```

```
[1] 4 9 10 6 5 7 7 6 7 5 5 11
```

```
[1] 6.833333
```

 **Intuition:** If we kept rolling **forever**, the running average would settle down and stop changing — that settling-down number is what we call the **expectation**.

Expectation

Suppose that we run a large number of random trials.

```
1 many_rolls <- replicate(1000, sum(sample(1:6, 2, replace = T)))  
2 mean(many_rolls)
```

```
[1] 7.064
```

The **expected value** or **expectation** is given by the **average** of all trials as the number of trials goes to infinity.

DEFINITION

Formally, the expectation of a (discrete) random variable X is given by

$$\mathbb{E}[X] = \sum_{x \in \mathcal{X}} x \cdot \mathbb{P}(X = x).$$

where $\mathcal{X}$ is the **support** of X .

In this course, we can think of the expectation of a random variable X is given by the **limit of the sample mean**

$$\mathbb{E}[X] = \lim_{n \rightarrow \infty} \left(\frac{\sum_{i=1}^n x_i}{n} \right).$$

Example — Expectation of a Uniform RV

Recall $X \sim \mathcal{U}(6)$ (one fair die roll) has PMF $p_X(k) = \frac{1}{6}$ for $k \in \{1, \dots, 6\}$.

Applying the definition of expectation:

$$\mathbb{E}[X] = \sum_{k=1}^6 k \cdot \frac{1}{6} = \frac{1 + 2 + 3 + 4 + 5 + 6}{6} = 3.5$$

We can confirm this by simulating many die rolls and averaging:

```
1 set.seed(1)
2 many_die_rolls <- replicate(10000, sample(1:6, size = 1))
3 mean(many_die_rolls)

[1] 3.5138
```

 In general, for $X \sim \mathcal{U}(n)$, $\mathbb{E}[X] = \frac{n+1}{2}$.



Exercise — Expectation

Suppose we **repeatedly flip two fair coins**. We are interested in determining the expected number of flips it takes to get both coins to land on tails.

1. Simulate 100 trials (one trial is flipping 2 coins).
2. Count the number of trials it takes to get two tails.
3. Record that number.
4. Repeat this experiment 1000 times.

Write a function using `replicate()` to calculate the number of trials until a double-tails flip, then use `replicate()` again for the 1000 experiments.

Again, you may find the function `which()` to be helpful.

04:00



Solution — Expectation

```
1 tt <- function() {  
2   flips <- replicate(  
3     100, sum(sample(0:1, 2, replace = T))  
4   )  
5   return(which(flips == 2)[1])  
6 }  
7 expected_flips <- replicate(1000, tt())  
8 mean(expected_flips)
```

```
[1] 3.967
```

- `tt()` flips two coins 100 times — `sum(sample(0:1, 2))` equals 2 exactly when **both land tails** — and returns the **index of the first double-tails flip** using `which(flips == 2)[1]`.
- `replicate(1000, tt())` repeats that experiment 1000 times, giving 1000 simulated **waiting times** until a double-tails result.
- `mean(expected_flips)` averages those waiting times to approximate the **expected number of flips**.



Bernoulli Distribution

DEFINITION

A **Bernoulli random variable** $X \sim \text{Bernoulli}(p)$ describes the outcome of an experiment with a **binary outcome** (a **Bernoulli trial**) where the probability of a “successful outcome” is p and the probability of failure is $1 - p$.

- A Bernoulli random variable only ever takes **two possible values**: $X = 1$ (success) or $X = 0$ (failure).
- E.g. a single coin flip, whether a patient recovers, or whether an email is spam.

The **probability mass function** of X is given by

$$p_X(k) = p^k(1 - p)^{1-k}; \quad k \in \{0, 1\}.$$

Plugging in $k = 0$ and $k = 1$ shows why this formula makes sense:

$$p_X(0) = p^0(1 - p)^{1-0} = 1 - p, \quad p_X(1) = p^1(1 - p)^{1-1} = p.$$



Example — Bernoulli Distribution

Back to coin flipping...

- Consider flipping a **biased coin** that is three times as likely to land on heads than on tails.
- Let X describe this experiment where $X = 1$ means heads and $X = 0$ means tails.
- The probability mass function of this Bernoulli random variable is $p_X(k) = 0.75^k \cdot 0.25^{1-k}$ for $k \in \{0, 1\}$.
- This gives the probabilities $\mathbb{P}(X = 1) = 0.75$ and $\mathbb{P}(X = 0) = 0.25$.

Binomial Distribution

DEFINITION

A **Binomial random variable** $X \sim \text{Bin}(n, p)$ describes the experiment of **counting the number of successes** in n independent Bernoulli trials, each with success probability p .

- Intuitively, a Binomial random variable is just a **sum of n independent Bernoulli(p) trials**, i.e. $X = X_1 + X_2 + \dots + X_n$ where each $X_i \sim \text{Bernoulli}(p)$.
- Each X_i contributes a 1 if that trial succeeds and 0 otherwise, so X simply **counts up the successes**.

The **probability mass function** of X is given by

$$\mathbb{P}(X = k) = \binom{n}{k} p^k (1 - p)^{n-k}; \quad k \in \{0, 1, \dots, n\},$$

where the **support** is all integers from 0 (no successes) to n (all successes).

12 34 Choose Function

In the PMF we used the **choose function** $\binom{n}{k}$ — n choose k — which is defined as

$$\binom{n}{k} = \frac{n!}{k!(n-k)!}.$$

This definition uses **factorials** where

$$n! = \prod_{i=1}^n i = 1 \times 2 \times \dots \times n.$$

 Combinatorially, $\binom{n}{k}$ counts the **number of ways to choose k items from a set of n** , without regard to order.

We can use the built-in function `choose(n, k)` to compute this in R.

Example — Binomial Distribution

Consider an experiment where we **flip the biased coin** from our Bernoulli example 12 times. We wish to determine the probability that we obtain 8 heads.

We define the random variable X , which describes the **number of heads** from the 12 throws. Therefore $X \sim \text{Bin}(12, 0.75)$ with support $\{0, 1, \dots, 12\}$, which gives

$$\mathbb{P}(X = 8) = \binom{12}{8} \cdot (0.75)^8 \cdot (0.25)^4.$$

Using the `choose(n=12, k=8)` function we find:

```
1 choose(12, 8) * (0.75)^8 * (0.25)^4
```

```
[1] 0.1935777
```

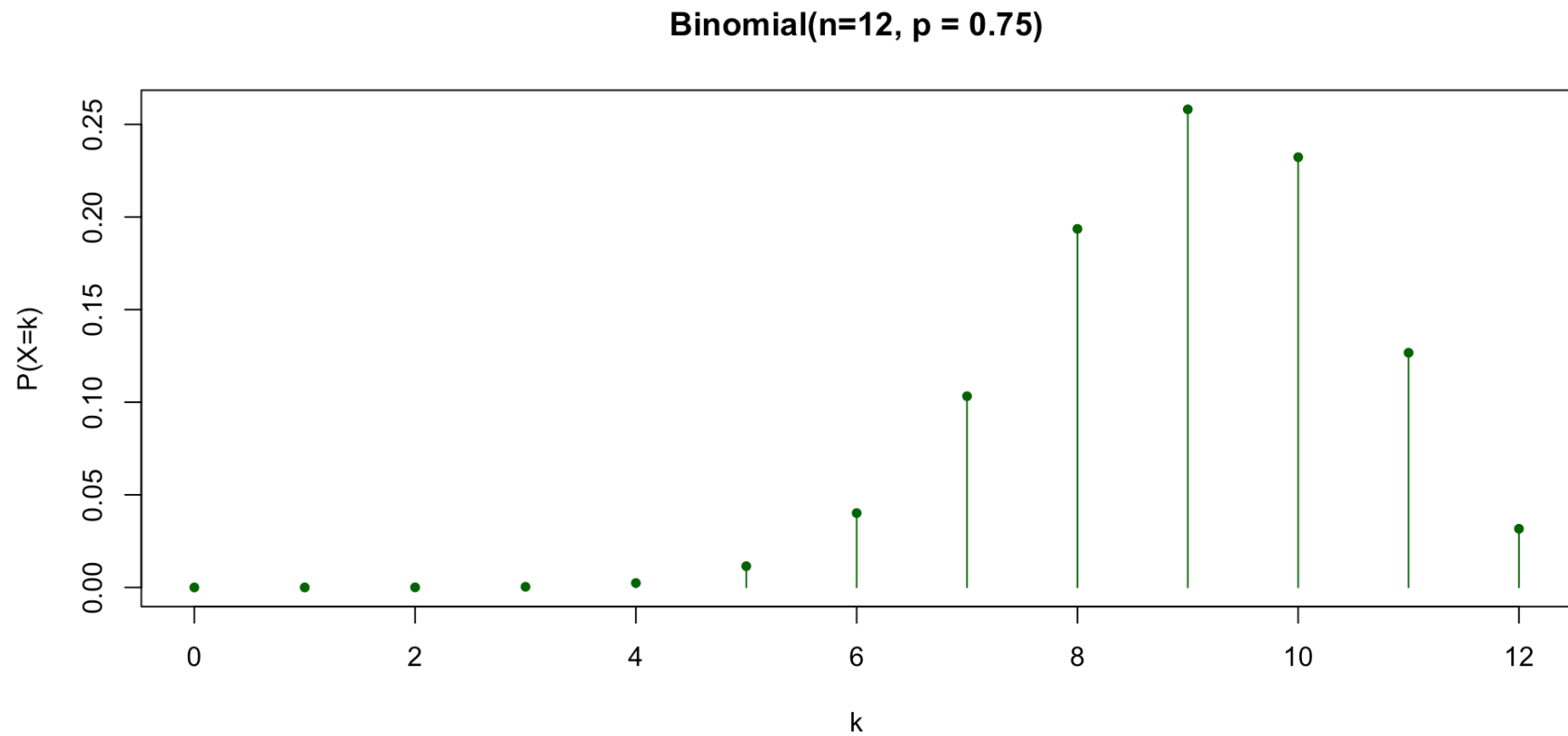
A couple more quick examples with different n , k and p values:

```
1 # Bin(n=10, p=0.5): P(X=4)
2 choose(10, 4) * (0.5)^4 * (0.5)^6
3 # Bin(n=20, p=0.3): P(X=5)
4 choose(20, 5) * (0.3)^5 * (0.7)^15
```

```
[1] 0.2050781
```

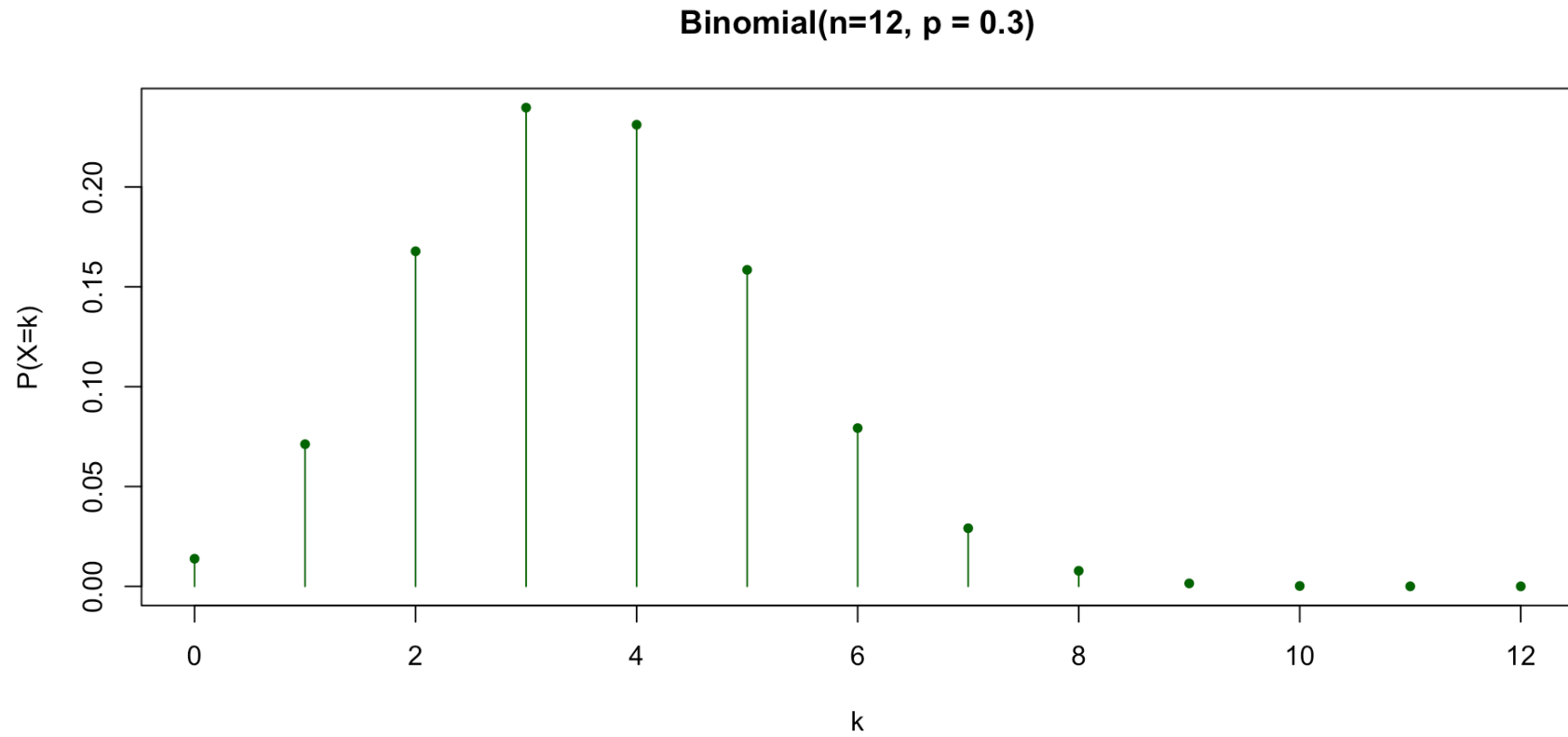
```
[1] 0.1788631
```

Binomial Distribution — $p = 0.75$



Binomial($n=12$, $p = 0.75$) Probability Mass Function

Binomial Distribution — $p = 0.3$



Binomial($n=12$, $p = 0.3$) Probability Mass Function

Example — Expectation of a Binomial RV

Since $X \sim \text{Bin}(n, p)$ is the sum of n independent Bernoulli(p) trials, and each trial succeeds with probability p on average, the expected number of successes is simply

$$\mathbb{E}[X] = np.$$

Recall $X \sim \text{Bin}(12, 0.75)$ from our biased-coin example. Applying the formula:

$$\mathbb{E}[X] = 12 \times 0.75 = 9.$$

We can confirm this by simulating many sets of 12 biased-coin flips and averaging the number of heads:

```
1 set.seed(1)
2 many_binom_trials <- replicate(10000, sum(sample(0:1, 12, replace = TRUE, prob = c(0.25, 0.75))))
3 mean(many_binom_trials)
```

```
[1] 9.0022
```



In general, for $X \sim \text{Bin}(n, p)$, $\mathbb{E}[X] = np$.



Cumulative Distribution Functions

DEFINITION

The **cumulative distribution function (CDF)** of a random variable X , denoted $F_X(x)$, describes the probability that X takes a value **less than or equal to** some value x , i.e.

$$F_X(x) = \mathbb{P}(X \leq x).$$

For a discrete random variable this probability is just the **sum of the probabilities** of the random variable taking the values **below** x , i.e.

$$F_X(x) = \sum_{k \leq x} p_X(k).$$

- The **PMF** gives the probability of hitting one **specific** value; the **CDF** accumulates (adds up) the PMF over all values up to x .
- Consequently the CDF is **non-decreasing** in x , and $\mathbb{P}(X = k) = F_X(k) - F_X(k - 1)$ recovers the PMF from the CDF.



Example — Binomial CDF

Returning to our binomial random variable example, let's look at **computing the CDF**, specifically $F_X(2)$, which is equivalent to

$$\begin{aligned} F_X(2) &= \mathbb{P}(X \leq 2) \\ &= \mathbb{P}(X = 0) + \mathbb{P}(X = 1) + \mathbb{P}(X = 2) \\ &= \binom{12}{0} \cdot 0.75^0 \cdot 0.25^{12} + \binom{12}{1} \cdot 0.75^1 \cdot 0.25^{11} + \\ &\quad \binom{12}{2} \cdot 0.75^2 \cdot 0.25^{10} \\ &= \frac{12!}{0! 12!} \cdot 0.75^0 \cdot 0.25^{12} + \frac{12!}{1! 11!} \cdot 0.75^1 \cdot 0.25^{11} + \\ &\quad \frac{12!}{2! 10!} \cdot 0.75^2 \cdot 0.25^{10} \\ &\approx 0.0000376. \end{aligned}$$



Example — Binomial CDF in R

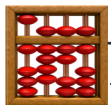
We can quickly compute this in R.

```
1 # define function
2 binomial_cdf <- function(n = 10, p = 0.5, k=5) {
3   result <- 0
4   for(i in 0:k){
5     result <- result + choose(n, i) * p^i * (1-p)^(n-i)
6   }
7   return(result)
8 }
9 binomial_cdf(n = 12, p = 0.75, k = 2)
```

```
[1] 3.761053e-05
```



Writing our own `binomial_cdf()` function works, but R actually has a **built-in function** for this — we'll meet it next.



Binomial R Functions

R has several **in-built functions** for computing and generating realizations from probability mass functions:

1. `dbinom()`

- Computes the **PMF**: $\mathbb{P}(X = k)$.

2. `pbinom()`

- Computes the **CDF**: $\mathbb{P}(X \leq k)$.

3. `qbinom()`

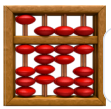
- Computes **quantiles** — the inverse of the CDF.

4. `rbinom()`

- **Generates random realizations** of the random variable.



This `d` / `p` / `q` / `r` naming pattern is **consistent across every distribution family** in R — e.g. `dnorm()`, `dpois()` and `dunif()` all follow the same syntax.



dbinom() Function

The `dbinom()` function computes the density of a binomial distribution.

The general syntax is:

```
1 dbinom(x = k, size = n, prob = p, log = FALSE)
```

Example:

For our $X \sim \text{Bin}(n = 12, p = 0.75)$ we compute $\mathbb{P}(X = 8)$:

```
1 choose(12, 8) * (0.75^8) * (0.25^4)
2 dbinom(8, size = 12, prob = 0.75)
```

```
[1] 0.1935777
```

```
[1] 0.1935777
```

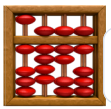
Another Example:

For $X \sim \text{Bin}(n = 20, p = 0.4)$ we compute $\mathbb{P}(X = 6)$:

```
1 choose(20, 6) * (0.4^6) * (0.6^14)
2 dbinom(6, size = 20, prob = 0.4)
```

```
[1] 0.1244117
```

```
[1] 0.1244117
```



`pbinom()` Function

The `pbinom()` function computes the cumulative distribution function of a binomial distribution.

The general syntax is the same:

```
1 pbinom(x = k, size = n, prob = p, log = FALSE)
```

Example:

For our $X \sim \text{Bin}(n = 12, p = 0.75)$ we compute $\mathbb{P}(X \leq 2)$:

```
1 binomial_cdf(n = 12, p = 0.75, k = 2)
2 pbinom(2, size = 12, prob = 0.75, lower.tail = TRUE)
```

```
[1] 3.761053e-05
```

```
[1] 3.761053e-05
```

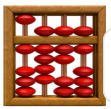
Another Example:

For $X \sim \text{Bin}(n = 20, p = 0.4)$ we compute $\mathbb{P}(X \leq 6)$:

```
1 binomial_cdf(n = 20, p = 0.4, k = 6)
2 pbinom(6, size = 20, prob = 0.4, lower.tail = TRUE)
```

```
[1] 0.2500107
```

```
[1] 0.2500107
```



`rbinom()` Function

The `rbinom()` function generates realizations of a binomial random variable.

The general syntax is the same:

```
1 rbinom(n = n_sims, size = n, prob = p, log = FALSE)
```

Example:

To simulate 10000 realizations of $X \sim \text{Bin}(n = 12, p = 0.75)$ and produce a histogram we have:

```
1 binomial_simulations <- rbinom(n = 10000, size = 12, prob = 0.75)
2 hist(
3   binomial_simulations,
4   main = "Binomial(12,0.75) simulations"
5 )
```

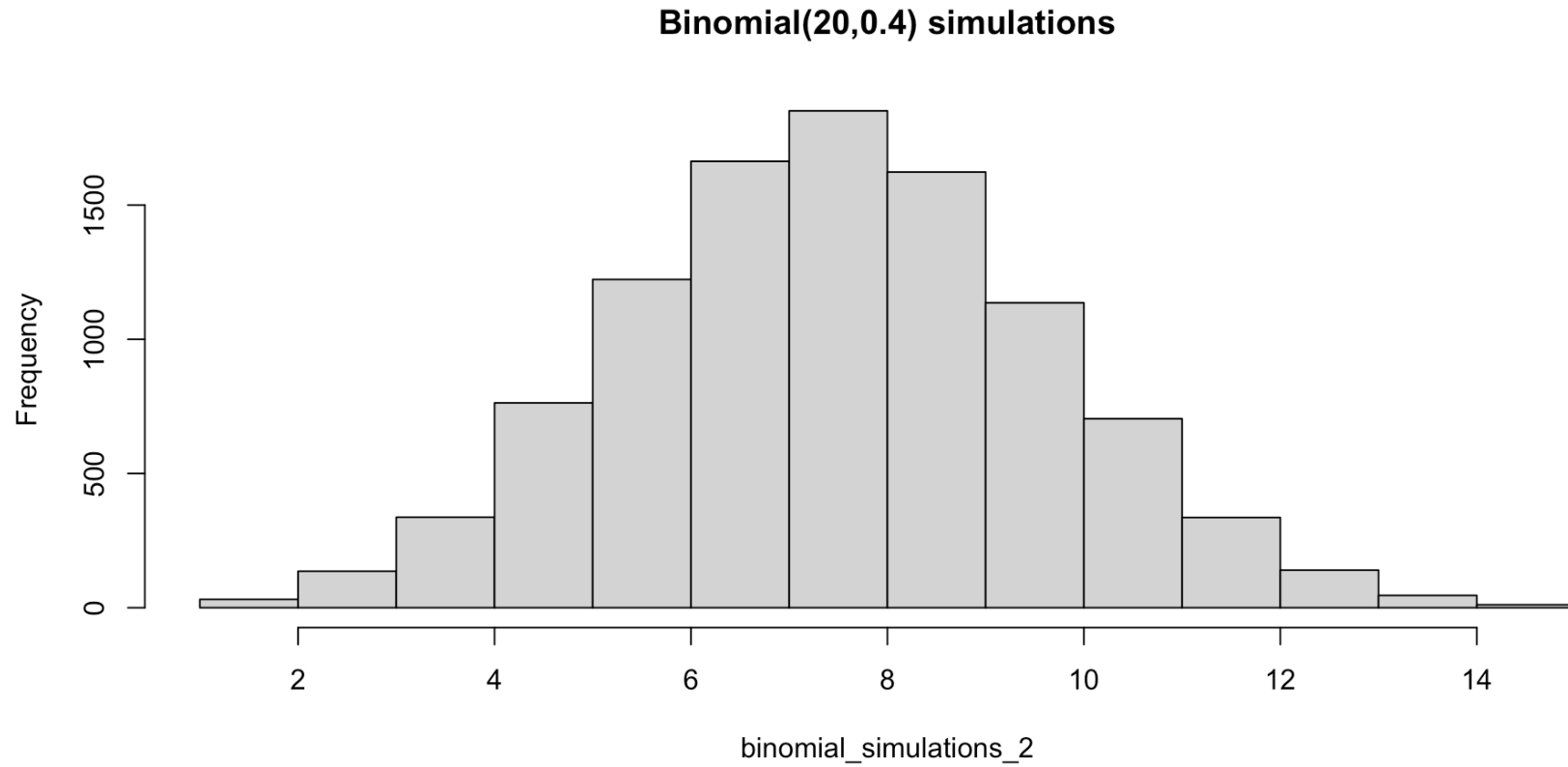
Another Example:

To simulate 10000 realizations of $X \sim \text{Bin}(n = 20, p = 0.4)$ and produce a histogram we have:

```
1 binomial_simulations_2 <- rbinom(n = 10000, size = 20, prob = 0.4)
2 hist(
3   binomial_simulations_2,
4   main = "Binomial(20,0.4) simulations"
5 )
```

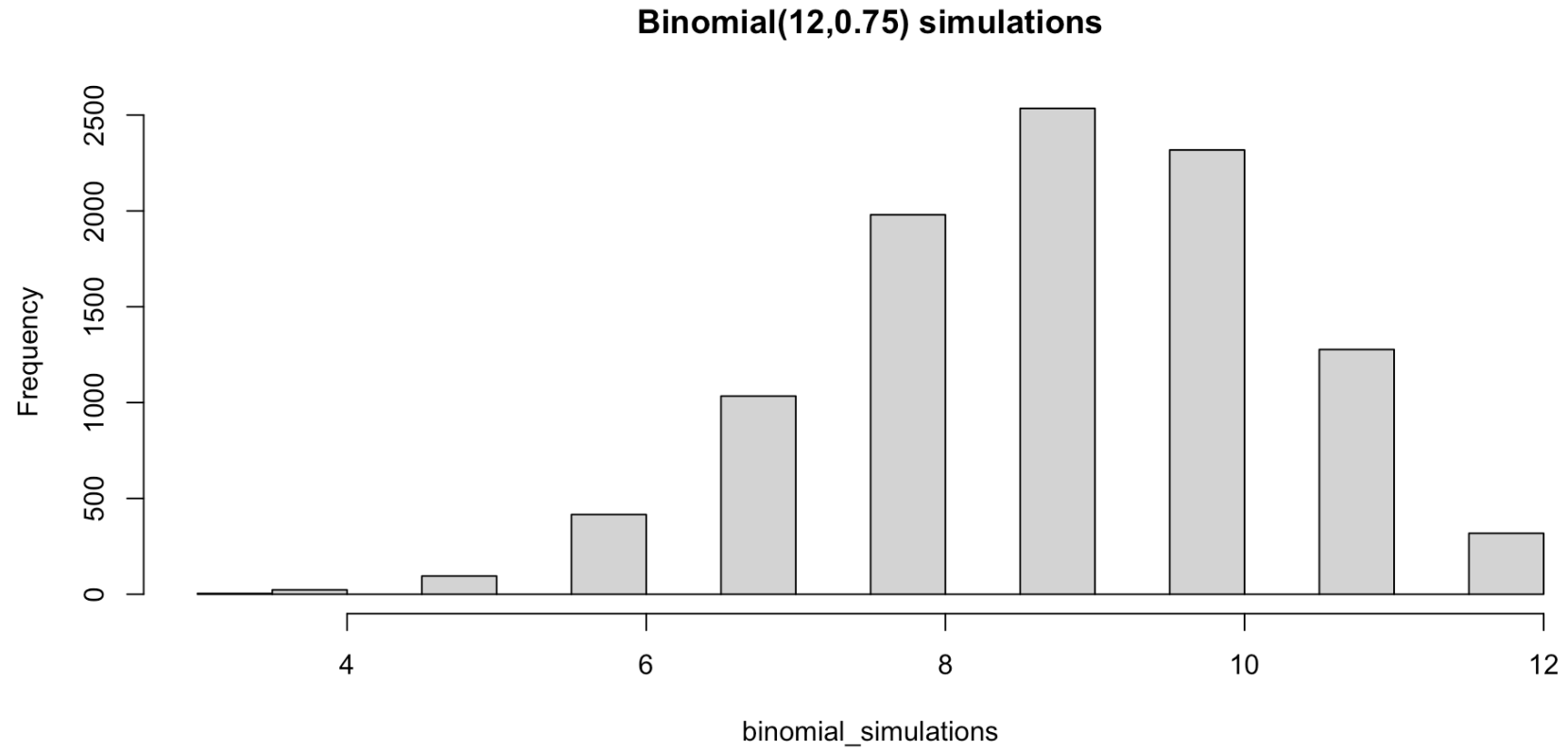


`rbinom()` Function





`rbinom()` Function



Summary

Topics Covered

 Today we covered a lot of material, including:

- Random Variables
- Probability Mass Functions
- Cumulative Distribution Functions
- Binomial Distributions
- Binomial R Functions:
 - `dbinom()`
 - `pbinom()`; and
 - `rbinom()`



Next Class



Next class we will continue exploring probability by looking into:

- Poisson Distributions
- Generalizing R Functions
- Continuous Random Variables
- Continuous Uniform Distribution
- Normal (Gaussian) Distribution